SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyze the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL2)*

QUESTIONS: 20 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I SIVA SUBRAMANIYAM B/192572089 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1. A college wants to identify students who may fail in the final exam. How can Data Science help?
2. A hospital has patient records for many years. How can Data Science be applied?
3. An online shopping website wants to recommend products to customers. Which use of Data Science is applied here?
4. A bank wants to detect suspicious transactions. How can Data Science be useful?
5. A company wants to reduce customer loss. How can Data Science support this?
6. A school wants to improve student performance. Mention one benefit of Data Science in this case.
7. A supermarket wants to know which products sell more during festivals. How can Data Science help?
8. A transport department wants to reduce traffic congestion. How can Data Science be applied?
9. A farmer wants to improve crop yield. Give one use of Data Science.
10. A company has data in Excel tables, customer reviews, images, and videos. Which facet of data is shown here?
11. A weather app updates temperature every minute. Which facet of data is involved?
12. A university stores student details in rows and columns. What type of data structure is this?
13. Customer comments on social media are collected for analysis. What type of data is this?
14. A company collects a very large amount of daily user activity data from its mobile app. Which Big Data characteristic is applied?
15. A hospital dataset contains missing age values and incorrect phone numbers. Which data quality issue is present?
16. A business wants to convert customer data into useful sales decisions. Which Big Data characteristic is important?
17. A company stores large files in Hadoop Distributed File System. Which part of the Big Data ecosystem is used?
18. A company wants to process large datasets using parallel computation. Which Big Data tool can be used?
19. An organization wants to collect live sensor data from IoT devices. Which Big Data ecosystem component is suitable?
20. A company wants to visualize Big Data results for decision-makers. Which tool or method can be applied?

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

ANSWERS:

1. Uses predictive analytics on past marks and attendance to flag at-risk students for early intervention.
2. Employs historical clinical analytics to forecast disease trends and design personalized patient care plans.
3. Recommendation systems (collaborative and content-based filtering based on user activity).
4. Uses real-time machine learning anomaly detection to flag transactions that deviate from normal spending habits.
5. Applies customer churn prediction to detect dissatisfaction early and send targeted retention offers.
6. Enables personalized learning by identifying each student's specific weaknesses to tailor instruction.
7. Employs demand forecasting and market basket analysis on past sales trends to optimize seasonal stock.
8. Analyzes real-time GPS and camera feeds using stream processing to dynamically optimize traffic signal timing.
9. Enables precision agriculture by evaluating soil and weather data to optimize irrigation and fertilizer usage.
10. Variety (the coexistence of structured, semi-structured, and unstructured data types).
11. Velocity (the high-speed, continuous generation and streaming of live data).
12. Structured data (relational tabular data organized into rows and columns).
13. Unstructured data (free-form text lacking a predefined schema).
14. Volume (the massive quantity and scale of data generated daily).
15. Data incompleteness and inaccuracy (a data veracity / data quality issue).
16. Value (transforming raw data into actionable, revenue-driving business decisions).
17. Distributed storage layer (Hadoop Distributed File System / HDFS).
18. Apache Spark (or Apache Hadoop MapReduce).
19. Apache Kafka (distributed real-time data ingestion and messaging system).
20. Business intelligence tools like Tableau or Power BI (interactive visual dashboards).